

Question:
There exist real numbers x and y , both greater than 1, such that $\log_x(y^x) = \log_y(x^{4y}) = 10$. Find xy .

Reasoning + Answer:

We are given :

$$\log_x(y^x) = 10 \quad \text{and} \quad \log_y(x^{4y}) = 10$$

We are to find xy , where $x > 1$, $y > 1$, and both are real numbers .

Step 1 : Simplify the first equation

$$\log_x(y^x) = 10$$

Use the logarithmic identity : $\log_b(a^n) = n \log_b a$

So :

$$\log_x(y^x) = x \log_x y = 10$$

Therefore :

$$x \log_x y = 10 \quad \text{(Equation 1)}$$

Step 2 : Simplify the second equation

$$\log_y(x^{4y}) = 10$$

Apply the power rule :

$$4y \log_y x = 10$$

So :

$$4y \log_y x = 10 \quad \text{(Equation 2)}$$

Step 3 : Use change of base

Note that $\log_x y = \frac{1}{\log_y x}$

Let us define :

Let $a = \log_x y$. Then $\log_y x = \frac{1}{a}$

Now substitute into the equations .

From Equation 1 :

$$x a = 10 \quad \Rightarrow \quad a = \frac{10}{x} \quad \text{(1)}$$

From Equation 2 :

$$4y \cdot \frac{1}{a} = 10 \quad \Rightarrow \quad \frac{4y}{a} = 10 \quad \text{(2)}$$

Now substitute $a = \frac{10}{x}$ into equation (2) :

$$\frac{4y}{10/x} = 10$$

Simplify :

$$\begin{aligned} 4y \cdot \frac{x}{10} &= 10 \\ \Rightarrow \frac{4xy}{10} &= 10 \\ \Rightarrow \frac{2xy}{5} &= 10 \end{aligned}$$

Multiply both sides by 5 :

$$\begin{aligned} 2xy &= 50 \\ \Rightarrow xy &= 25 \end{aligned}$$

<answer>25</answer>